A

REPORT

of

Mini Project Or Internship Assessment

At

“S O Infotech (P) Ltd.”

on

“ Pulse – Real-time Analytics Dashboard ”

Submitted

In the partial fulfilment of Bachelor of Technology Department of Computer Science and Engineering

Shri Ram Murti Smarak College of Engineering & Technology, Bareilly

Session: 2026-27

Submitted by:

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(2300140100012)

Submitted to:

Dr. Juhi Chaudhary

(Assistant Professor CSE Department)

CERTIFICATE (By Company)

S O INFOTECH (P) LTD.
PULSATING TECHNOLOGIES

Dated:19/07/2026 Ref No. SO/PL: 26/18310

TO WHOM IT MAY CONCERN

This is to certify that **Ms. AMBIKA GUPTA** S/O or D/O **Mr. ANIL GUPTA, B.TECH,** student from. **SHRI RAM MURTI SMARAK COLLEGE OF ENGINEERING & TECHNOLOGY** has successfully completed the project at our organization from **JUNE 2026 TO JULY 2026,** under training period, he/she worked on the project entitle **"REAL TIME ANALYTICS DASHBOARD"** as **PYTHON DEVELOPER** under the guidance of our project manager **Mr. YASIR.**

During this training he/she was an active member of the team working on the project. The work carried by him/her was satisfactory and wish him all the best for his/her future assignments.

For S. O infotech Pvt. ltd.

Authorized Signatory
Best Regards,
Auth. Signatory

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DECLARATION

This is to certify that Ambika Gupta**,** student of Shri Ram Murti Smarak College of Engineering and Technology, Bareilly, of branch Computer Science bearing Roll No. 2300140100012 has successfully completed Internship in Python Development Intern to my satisfaction and submitted the same during the academic year 2026-27 towards the partial fulfilment of final year of B. Tech under the Department of Computer Science, SRMS-CET,
Bareilly.

Signature: _____

Name: Ambika Gupta

Roll no: 2300140100012

ACKNOWLEDGEMENT

I would like to express my sincere gratitude and appreciation to all those who have contributed to the successful completion of my internship at S O INFOTECH (P) Ltd. This experience has been invaluable in enhancing my skills and knowledge in the field of Python Development.

I am thankful to the entire team at S O INFOTECH for providing me with an environment that fostered learning and growth. The collaborative atmosphere and exposure to real-world projects have greatly contributed to my professional development. I appreciate the opportunities I had to work alongside skilled professionals who willingly shared their knowledge and expertise.

In conclusion, my internship experience at S O INFOTECH (P) Ltd. has been transformative, and I am thankful for the opportunities and knowledge I have gained. I look forward to apply the skills I've acquired in my future endeavors and contributing positively to the field of Python Development.

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Signature: _____

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CHAPTER 1

COMPANY PROFILE

S O Infotech follows a functionally organized structure designed to support efficient project execution and client servicing. At a high level, the organization is structured around core functional units, including software development teams, quality assurance and testing teams, technical training and resourcing units, and client-facing project coordination roles.

1.1. Company Mission and Vision

S O Infotech (P) Ltd. operates with the mission of delivering cost-effective, high-quality software and IT solutions that help client businesses maximize the value of their technology investments. As a software development outsourcing company with a strong focus on e-finance, e-business, and media domains, the organization aims to bridge the gap between business requirements and technical execution through reliable, scalable, and innovative solutions.

The company's vision is to be recognized as a globally benchmarked solution provider a preferred technology partner that creates high value for its customers while fostering an enriching workplace where employees can excel through innovation and teamwork. This vision is supported by a consistent commitment to delivering high-fidelity services and solutions, underpinned by time-tested, institutionalized processes built on the belief that quality is ultimately a cost-saving factor rather than an added expense. By combining a strong business-to-IT connect with deeply committed people, S O Infotech seeks to give its clients a genuine and sustained competitive edge in their respective industries.

1.2. Core Values

The organizational culture at S O Infotech is anchored in a well-defined set of core values that guide day-to-day operations, client engagements, and internal collaboration:

a) Agility - the ability to respond quickly and effectively to changing client needs and market conditions.

b) Dependability - building trust through consistent, reliable delivery on commitments.

c) Innovation - encouraging creative problem-solving and the adoption of modern technologies and practices.

d) Integrity - maintaining honesty and transparency in all business and interpersonal dealings.

e) Meritocracy & Fair Play - recognizing and rewarding contribution and capability rather than tenure alone.

f) Passion - approaching work with genuine enthusiasm and commitment to excellence.

g) Teamwork - fostering collaboration across teams and departments to achieve shared goals.

These values collectively shape a work environment where employees are encouraged to take ownership of their work while contributing to a larger, shared organizational purpose.

1.3. Products and Services

S O Infotech offers a diverse portfolio of services tailored to meet the technology needs

of businesses across multiple industries. The company's core service offerings include:

1.3.1. Website Development

a) Custom, responsive website design and development

b) Tailored to client branding and functional requirements.

1.3.2. Industrial Training

a) Structured, hands-on technical training for students and early-career professionals

b) This internship was completed under this initiative.

1.3.3. Advanced Programming & Application Development

a) Custom software and enterprise-grade application development.

b) Built using modern programming framework.

1.3.4. Quality Assurance Services

a) Structured functional and performance testing.

b) Ensures reliability of delivered software

1.3.5. SEO/SEM & Web Optimization

a) Search engine optimization and marketing strategies

b) Improves visibility, ranking, and performance of client platforms

1.3.6. Support and Maintenance

a) Offering dedicated helpdesk and remote assistance for deployed solutions.

b) Regular updates, patches, and performance optimization for all deployed software systems.

1.4. Organizational Structure

S O Infotech follows a functionally organized structure designed to support efficient project execution and client servicing. At a high level, the organization is structured around core functional units, including software development teams, quality assurance and testing teams, technical training and resourcing units, and client-facing project coordination roles.

a) Software Development Department: Responsible for designing, developing, and maintaining software applications, including ERP systems, institutional management tools, and business automation solutions.

b) Training and Skill Development Department: Delivers industry-oriented technical training, internships, and certification programs in technologies like Java, Python, PHP, .NET, IoT, and Machine Learning.

c) Workshops and Seminars Department: Organizes hands-on workshops, webinars, and seminars to enhance technical knowledge and soft skills of students and professionals.

d) Academic Solutions Department: Specializes in developing and implementing Learning Management Systems (LMS), online examination platforms, and attendance tracking solutions for institutions.

e) Industry-Specific Solutions Department: Creates tailored IT solutions for sectors such as manufacturing, pharmaceuticals, and healthcare.

f) Web and Mobile Development Department: Designs and develops secure, user-friendly, and responsive web portals and mobile applications for various clients.

1.5. Company Culture and Work Environment

S O Infotech (P) Ltd. fosters a collaborative and innovative work environment.The company values continuous learning and professional development encouraging employees to stay updated with the latest technologies and industry trends. The culture emphasizes teamwork, creativity, and a strong commitment to delivering exceptional results for clients.

1.5.1. **Employee Benefits**:

- Professional Development: Opportunities for training, certifications, and career advancement.
- Work-Life Balance: Flexible work arrangements and supportive policies to maintain a healthy work-life balance.
- Innovative Environment: A culture that encourages creativity , experimentation, and innovation.

1.5.2. Clientele and Market Presence

S O Infotech Pvt. Ltd. serves a diverse clientele across various industries, including healthcare, finance, retail, and more. The company's strong focus on delivering customized solutions and its ability to adapt to different industry needs have contributed to its success and growth it's reputation for excellence and its commitment to client satisfaction have earned it a strong market presence and a loyal customer base.

1.6. Future Outlook

Looking ahead, S O Infotech continues to focus on strengthening its position as a reliable, innovation-driven IT services provider. As client demand increasingly shifts toward real-time systems, data-driven decision-making, and scalable cloud-based architectures, the company is positioned to expand its expertise in these areas building on its existing strengths in web development, application engineering, and quality assurance.

Continued investment in industrial training programs, such as the one through which this project was completed, also reflects the company's long-term strategy of cultivating a pipeline of technically skilled talent, while simultaneously exposing students to industry-relevant, production-style software development practices. This dual focus on client-facing service delivery and talent development positions S O Infotech to sustain its growth while contributing meaningfully to the broader technology education ecosystem.

CHAPTER 2

INDUSTRY AT A GLANCE

The technology industry is one of the most dynamic and rapidly evolving sectors globally, spanning disciplines from software development and web engineering to real-time systems and data analytics. As businesses increasingly rely on technology to optimize operations, engage customers, and make faster, data-driven decisions, the demand for scalable, real-time software solutions and skilled professionals continues to grow. This chapter provides an overview of the core service areas within the IT industry, with particular reference to how S O Infotech (P) Ltd. operates within each.

2.1 Software Development

2.1.1 Custom Software Development:

S O Infotech specializes in creating tailored software solutions to meet the unique requirements of its clients. Its custom development services span the full lifecycle from initial requirement gathering and system design through development, testing, and deployment ensuring that each solution is purpose-built rather than adapted from a generic template.

2.1.2 Web Application Development:

In web application development, S O Infotech builds robust, interactive applications covering both front-end and back-end components. The company works with modern frameworks and languages across the Python and JavaScript ecosystems, enabling the delivery of applications that are functionally reliable as well as responsive and user-friendly — an approach directly reflected in the backend and real-time dashboard development undertaken during this internship.

2.1.3 Backend and Real-time Systems:

With growing client demand for instant data visibility, S O Infotech has expanded its

capabilities into backend architectures that support real-time data processing. This includes API development, database design, and event-driven systems using technologies such as websockets and message-broker services, allowing applications to push live updates to users.

2.2. IT Consulting and Advisory

2.2.1. Technology Strategy and Planning:

S O Infotech provides strategic IT consulting services to help businesses align their technology investments with their overall goals and objectives. This involves assessing current technology infrastructure, identifying gaps, and recommending improvements. Their consultants work closely with clients to develop technology roadmaps and implementation plans that drive business growth and efficiency.

2.2.2. Digital Transformation Consulting:

In today's rapidly evolving digital landscape, businesses need to continuously adapt to stay competitive. S O Infotech offers digital transformation consulting to help organizations leverage new technologies to enhance their operations. This includes evaluating existing processes, identifying opportunities for automation and digital innovation, and guiding the implementation of cutting-edge solutions.

2.2.3. Cyber security Consulting:

Protecting sensitive data and maintaining secure IT environments are critical concerns for businesses. S O Infotech provides cybersecurity consulting services to assess and strengthen an organization's security posture. Their services include risk assessments, vulnerability testing, and the development of security policies and procedures to safeguard against cyber threats.

2.3. Enterprise Solutions

2.3.1 Data and Analytics Platforms:

S O Infotech develops platforms that help organizations consolidate and interpret operational data, ranging from periodic reporting tools to live analytics dashboards. These systems integrate data collection, storage, and visualization into a single, coherent platform — the same category of system this internship project belongs to.

2.3.2 Workflow and Process Automation:

The company also builds systems that automate routine business processes, reducing manual effort and minimizing errors. This includes automating data capture and notification workflows, conceptually similar to the signal-triggered, event-driven notification pipeline implemented in this project.

2.4.UI/UX Design

2.4.1 User Interface (UI) Design:

S O Infotech places a strong emphasis on creating intuitive and engaging user interfaces that seamlessly blend aesthetics with functionality. The company follows modern design principles, ensuring that each interface is visually appealing, easy to navigate, and consistent across devices and platforms. Their UI design process involves in-depth research into user preferences, competitor analysis, and industry trends to craft layouts that resonate with the target audience.

2.4.2 User Experience (UX) Design:

Alongside UI design, S O Infotech provides comprehensive UX design services aimed at delivering applications that offer a seamless, intuitive, and satisfying experience for users. The process begins with in-depth user research to understand the target audience's behavior, goals, pain points, and expectations. Insights from this research are translated into clear user flows, ensuring that every step a user takes within the application feels logical and effortless. The company develops wireframes and interactive prototypes to visualize the structure and functionality before full-scale development, enabling stakeholders to provide feedback early in the design phase. Usability testing is a critical part of their approach, where real users interact with the product to identify any friction points or areas for improvement. S O Infotech also applies accessibility best practices, ensuring that their solutions are inclusive and usable by individuals with diverse needs. The focus is not just on aesthetics but on crafting experiences that boost engagement, reduce errors, improve efficiency, and leave users with a positive impression that encourages long-term usage.

2.5 Support and Maintenance

2.5.1 Technical Support:

S O Infotech provides ongoing technical support to ensure that clients' IT systems and applications run smoothly. Their support services include troubleshooting, issue resolution, and regular maintenance to address any technical problems that may arise.

2.5.2 System Maintenance:

Regular maintenance is essential for keeping IT systems up-to-date and secure. S O Infotech offers system maintenance services to apply updates, patches, and upgrades to software and hardware, ensuring optimal performance and security.

2.6 Training and Development

2.6.1 Technical Training:

To empower clients with the skills needed to effectively use and manage their technology solutions, S O Infotech offers technical training programs. These programs cover various technologies and tools, providing hands-on experience and knowledge to clients' and various teams.

2.6.2 Mentorship-Based Skill Building:

Rather than purely theoretical instruction, training emphasizes hands-on problem solving, including debugging real technical issues — allowing trainees to build practical, job-ready skills alongside academic knowledge.

2.7 Industry-Specific Solutions

2.7.1 Health care IT Solutions:

S O Infotech delivers specialized IT solutions tailored to meet the unique requirements of the

healthcare industry, focusing on improving patient care, enhancing operational efficiency, and ensuring data security. Their offerings include the development and implementation of **Electronic Health Records (EHR)** systems that enable healthcare providers to securely store, access, and share patient data in real time. The company also designs **telemedicine platforms** that connect doctors and patients remotely, expanding access to healthcare services, especially in rural or underserved areas.

2.7.2 Finance and Banking Solutions:

S O Infotech provides robust and secure IT solutions for the finance and banking sector, designed to streamline operations, enhance customer experience, and ensure compliance with regulatory standards. Their offerings include the development of **core banking systems** that centralize and automate essential banking functions, enabling seamless management of accounts, transactions, and customer data across multiple branches

2.7.3 Retail Solutions:

S O Infotech offers innovative IT solutions for the retail industry, aimed at enhancing customer engagement, optimizing inventory management, and streamlining sales operations. Their **Point-of-Sale (POS) systems** are designed for speed, accuracy, and ease of use, enabling retailers to process transactions efficiently while integrating seamlessly with inventory and customer databases.

CHAPTER 3

TOOLS AND TECHNOLOGY UTILIZED IN INDUSTRY

Modern software development relies on a diverse ecosystem of tools and technologies, each addressing a specific stage of the development lifecycle - from writing and testing code to deploying scalable, data-driven applications. This chapter outlines the major categories of tools used across the industry, with particular reference to those directly applied during this internship project.

3.1 Software Development Tools

3.1.1. Integrated Development Environments (IDEs):

a) Visual Studio: A comprehensive IDE from Microsoft used for developing.NET applications, including web, desktop, and mobile apps.

b) IntelliJ IDEA: A powerful IDE for Mern development and other JVM-based languages, offering advanced code assistance and tools for building robust applications.

c) Eclipse: An open-source IDE commonly used for Mern development, butal so supports other languages through plugins.

3.1.2. Programming Languages:

a) MERN Stack: Essential for web development, MERN stack is used for creating interactive and dynamic webpages. Frameworks like React and Angular enhance its capabilities.

b) Html: HTML (HyperText Markup Language) is the standard language used to create and structure content on the web. It defines the basic structure of web pages using elements like headings, paragraphs, links, images, and forms.

c) CSS: CSS (Cascading Style Sheets) is used to style and layout web pages. It controls the appearance of HTML elements, including colors, fonts, and spacing.

3.1.3. Version Control System:

a) Git: A distributed version control system used to manage code changes, track revisions, and collaborate with other developers. Git repositories are often hosted on platforms like GitHub or GitLab.

3.1.4. Build and CI/CD Tools:

a) Jenkins: An open-source automation server used to build, test, and deploy applications continuously. Jenkins integrates with various tools and plugins to support DevOps practices.

b) Maven: A build automation tool primarily used for Mern projects, managing project dependencies and building project artifacts.

c) Docker: A containerization platform that allows developers to package applications and their dependencies into containers for consistent deployment across environments.

3.2 Web and Mobile Development Technologies

3.2.1. Web Development Frameworks:

a) React: A MERN stack library for building user interfaces, particularly single-page applications (SPAs), developed and maintained by Facebook.

b) Angular: A TypeScript-based framework developed by Google for building dynamic and scalable web applications.

c) Vue.js: A progressive MERN stack frame work used for building user interfaces and single-page applications with a focus on ease of integration and flexibility.

3.2.2 Mobile App Development Platforms:

a) Android Studio: The official IDE for Android development, providing tools for building, testing, and debugging Android applications.

b) Flutter: A UI toolkit from Google for building natively compiled applications for mobile, web, and desktop from a single codebase using the Dart language.

3.3 Cloud Solutions Technologies

3.3.1. Cloud Service Providers:

a) Amazon Web Services (AWS): A comprehensive cloud computing platform offering services such as computing power, storage, and databases. Key services include EC2 (virtual servers), S3 (object storage), and RDS (managed databases).

b) Microsoft Azure: A cloud platform from Microsoft providing a range of services including virtual machines, app services, and SQL databases. Azure also offers tools for AI, machine learning, and analytics.

c) Google Cloud Platform (GCP): Google's cloud computing services include Compute Engine (virtual machines), Cloud Storage, and BigQuery (data analytics).

3.3.2 Cloud Management and Monitoring Tools:

a) Kubernetes: An open-source platform for managing containerized applications across clusters of machines, providing automation for deployment, scaling, and operations.

b) Terraform: An infrastructure-as-code tool that allows users to define and provision cloud infrastructure using a declarative configuration language.

3.4 Data Analytics and Business Intelligence Tools

3.4.1 Data Analytics Tools:

a) Apache Spark: An open-source data processing engine for big data analytics, known foritsspeedandeaseofuse.Sparksupportsvariousdataprocessingtaskssuchasbatch processing and streaming.

b) Hadoop: A framework for distributed storage and processing of large data sets using a cluster of computers. It includes components like HDFS (Hadoop Distributed File System) and Map Reduce.

3.4.2 Business Intelligence (BI) Tools:

a) Power BI: A Microsoft tool for creating interactive visualizations and business intelligence reports, offering integration with various data sources.

b) Tableau: A data visualization tool that helps users create interactive and shareable dashboards, providing insights through intuitive and dynamic graphics.

3.5 Artificial Intelligence (AI)and Machine Learning (ML)Tools

3.5.1 Machine Learning Frameworks:

a) TensorFlow: TensorFlow is an open-source machine learning framework developed by Google, widely used for building, training, and deploying AI models.

b) PyTorch:. PyTorch is an open-source deep learning framework developed by Facebook's AI Research lab, known for its flexibility, ease of use, and strong support for dynamic computational graphs.

3.5.2 Computer Vision Tools:

a) OpenCV: An open-source computer vision library that provides tools for image and video analysis, including object detection and recognition.

b) Microsoft Azure Computer Vision: A cloud-based service offering image analysis, text extraction, and object detection capabilities.

3.6 User Interface (UI) and User Experience (UX) Design Tools

3.6.1. Design Tools:

a) Adobe XD:. Adobe XD is a powerful design and prototyping tool used to create user interfaces and user experiences for web, mobile, and desktop applications. It allows designers to craft wireframes, interactive prototypes, and high-fidelity UI designs in a collaborative environment.

b) Sketch:. Sketch is a vector-based design tool primarily used for creating user interfaces, website layouts, and mobile application designs. Known for its simplicity and lightweight performance, it provides designers with powerful features such as reusable symbols, shared styles, and an extensive library of plugins to streamline the design workflow.

3.6.2 Prototyping and Collaboration Tools:

a) Figma: Figma is a cloud-based interface design and prototyping tool that enables real-time collaboration among designers, developers, and stakeholders. It allows the creation of wireframes, high-fidelity UI designs, and interactive prototypes directly in the browser or through its desktop application, eliminating the need for complex installations

b) InVision:. InVision is a digital product design and prototyping platform used to create interactive mockups and collaborative design workflows. It allows designers to transform static screens into clickable, animated prototypes that simulate the final code.

CHAPTER 4

PROJECT SCOPE & SYSTEM DESCRIPTION

4.1. Introduction to the Project

Pulse is a Real-Time Analytics Dashboard developed to capture, process, and visualize user activity data as it occurs, rather than relying on periodic or manually refreshed reports. The project was undertaken as part of an industrial training internship at S O Infotech (P) Ltd., with the goal of applying academic knowledge of Python, SQL, and web development to a practical, production-style system. At its core, the project demonstrates how a modern web application can move data from the moment it is generated to a live, visual representation on a user's screen within a fraction of a second.

4.2 Problem Statement

Businesses that generate continuous user activity page views, clicks, signups, and purchases often lack a way to observe this activity as it happens. Conventional dashboards depend on the user manually refreshing the page or on scheduled batch reports, both of which introduce a delay between an event occurring and it becoming visible to decision-makers. This delay directly limits how quickly a business can respond to a sudden traffic spike, a drop in conversions, or a broken user flow. There is a clear need for a system that captures events instantly, processes them efficiently, and pushes updates to a dashboard the moment they happen without requiring any manual intervention from the person viewing the data.

4.3 Project Objectives

The project was guided by the following core objectives:

• To build a backend capable of capturing and processing user-activity events as they occur.

• To enable live visualization of activity data without requiring manual page refresh.

• To apply Pandas for converting raw event records into meaningful, aggregated analytics.

• To implement a genuine push-based architecture using WebSockets, rather than repeated polling of the server.

• To simulate realistic user traffic through a companion demo storefront, validating the complete pipeline end-to-end.

• To deliver a scalable, database-backed system that reflects patterns used in real production analytics platforms.

4.4 Scope of the Project

The scope of this project covers the design and implementation of a complete backend and frontend system for real-time event tracking and visualization. This includes: database design for storing event data; REST API development for data retrieval and submission; a real-time notification pipeline using Django Channels and Redis; a data-analysis layer using Pandas; and a browser-based dashboard interface with live charts and an event feed. The project also includes a demonstration storefront application, built specifically to generate realistic traffic and validate that the system correctly handles events originating from an independent client-facing source.

4.5 Out of the Scope

To maintain a focused and achievable project within the internship timeframe, certain areas were deliberately excluded. These include: user authentication and multi-tenant access control for dashboard viewers; production-grade deployment configuration (such as containerization or cloud hosting); advanced analytics such as predictive modelling or anomaly detection; and support for extremely high-throughput event volumes beyond what a single Redis instance can comfortably handle. These areas represent natural directions for future extension of the system rather.

4.6. System Overview

At a high level, the system consists of three cooperating layers. The data layer (PostgreSQL) permanently stores every event as a structured record. The processing and messaging layer (Django, Django Channels, and Redis) handles incoming requests, runs analytics via Pandas, and broadcasts new events to connected

clients using a publish/subscribe pattern. The presentation layer (the dashboard and demo storefront, built with HTML, CSS, JavaScript, and Chart.js) allows users to generate and observe activity in real time. These layers communicate through well-defined interfaces REST APIs for standard data exchange, and a persistent WebSocket connection for live updates keeping each layer independently testable and replaceable.

4.7. Functional Requirements

The system is expected to fulfil the following functional requirements:

• Capture user-activity events (page views, clicks, signups, purchases) and store them reliably in the database.

• Provide an API to retrieve recent events in raw form, for historical inspection.

• Provide an API to retrieve aggregated analytics (totals, breakdowns, top pages) computed using Pandas.

• Notify all connected dashboard clients immediately whenever a new event is recorded.

• Render a live-updating dashboard interface, including summary statistics, a chart, and a scrolling event feed.

• Provide a demo storefront capable of generating realistic events through normal user interaction.

4.8. Non-Functional Requirements

Beyond core functionality, the system was designed with the following quality attributes in mind:

• Low latency: updates should reach the dashboard within roughly one second of an event occurring.

• Reliability: event data must be saved to the database even if no dashboard is currently connected to view it.

• Scalability: the messaging layer (Redis) should support multiple simultaneous dashboard viewers without requiring each one to poll the database directly.

• Maintainability: each responsibility (data storage, API logic, real-time notification, presentation) should be isolated into its own module, allowing independent testing and modification.

4.9. Assumptions and Constraints

The project assumes that a single Redis instance and a single PostgreSQL database are sufficient for the expected scale of the demonstration environment, and that all components run on the same local network during development and testing. A notable technical constraint encountered during development was a compatibility issue between the Redis Python client library and Windows' asynchronous I/O handling, which required pinning a specific package version to resolve. The system also assumes that event data submitted through the tracking API is reasonably well-formed, as extensive input validation was considered outside the primary scope of this internship project.

4.10. Expected Outcome

The expected outcome of this project is a fully functional, end-to-end real-time analytics system: a user action on the demo storefront should result in a visible update on the dashboard reflected in the summary statistics, the live chart, and the event feed within roughly one second, without any manual refresh. Beyond the working system itself, the project is intended to demonstrate a practical understanding of real-time backend architecture, database-driven analytics, and the kind of production-style engineering decisions (technology selection, debugging, and system design trade-offs) expected in professional software development.

CHAPTER 5

CORE PROJECT ASSIGNED & MODULES IMPLEMENTED

The Pulse — Real-Time Analytics Dashboard system is built using a modular architecture, where each Django file or component handles a distinct, well-defined responsibility. This separation of concerns allows individual modules to be developed, tested, debugged, and extended independently, while still working together seamlessly to form the complete real-time data pipeline. Rather than a single monolithic file handling all logic, the system distributes responsibility across ten core modules, each described in detail below.

5.1 Data Model Module (models.py)

• **Functionality:** Defines the structure of the data captured by the system. This module represents single source of truth for what an "event" is within the application, and every other module ultimately reads from or writes to the structure defined here.

• **Components:**

(a) Event class - the core model, defining fields including user_id, event_type, page, timestamp, and an optional metadata field

(b) Field validation - the event_type field is restricted to a fixed set of choices (page_view, click, signup, purchase), preventing invalid or inconsistent data from entering the system

(c) Auto-timestamping - the timestamp field is automatically populated at the exact moment record is created, using auto_now_add=True

(d) String representation - a __str__ method defines how each event is displayed in the Django Admin panel, showing the user, event type, and time at a glance

Working: When any part of the application needs to create, read, or filter event data, it does so through this model, using Django's Object-Relational Mapper (ORM). This means

developers write Python code (e.g., Event.objects.create(...)) instead of raw SQL, while Django translates these operations into the appropriate PostgreSQL queries behind the scenes

5.2 API and View Module (views.py)

• **Functionality**: Handles all incoming HTTP requests and determines what data or page should be returned in response. This module acts as the primary interface between the frontend (browser) and the backend (database), and contains the business logic of the application.

• **Components:**

(a) event_list - returns the most recent 500 events as JSON, ordered by newest first, used to populate the Historical Data view

(b)analytics_summary - reads raw event data into a Pandas DataFrame and computes aggregated statistics: total event count, unique user count, event-type breakdown, and top pages by activity

(c) track_event - receives POST requests (typically from the demo storefront), parses the JSON request body, and creates a new Event record

(d) dashboard_page - renders and returns the main dashboard HTML template

(e) demo_site - renders and returns the demo storefront HTML template

Working: Each function in this module corresponds to exactly one URL, and Django automatically calls the matching function whenever a request arrives at that URL. Functions returning data (like event_list and analytics_summary) use JsonResponse, while functions returning full pages (like dashboard_page) use Django's render() function to combine an HTML template with any needed context data.

5.3 URL Routing Module (urls.py)

• **Functionality:** Maps incoming request URLs to the correct view function, acting as the entry point that directs traffic to the appropriate part of the application. Without this module, Django would have no way of knowing which function should handle a given web request.

• **Components:**

(a) /api/events/ - routes to event_list, the raw data-retrieval endpoint

(b) /api/summary/ - routes to analytics_summary, the aggregated-analytics endpoint

(c) /api/track/ - routes to track_event, the event-submission endpoint used by the demo storefront

(d) /api/dashboard/ - routes to dashboard_page, serving the live dashboard interface

(e) /api/demo/ - routes to demo_site, serving the demo storefront interface

Working: Django checks each incoming request's URL path against the list of patterns defined in this file, top to bottom, and calls the first matching view. This project's URL patterns are further included into the main project-level urls.py, keeping application-specific routes cleanly separated from project-wide configuration.

5.4 Admin Management Module (admin.py)

• **Functionality**: Provides a ready-made administrative interface for viewing, filtering, and managing stored data directly through a browser, without requiring custom queries or a separate database management tool.

• **Components:**

(a) EventAdmin class - registers the Event model with Django's built-in admin site

(b) list_display configuration - determines which fields (user, event type, page, timestamp) are visible in the admin's list view

(c) list_filter configuration - allows quick filtering of events by type directly from the admin sidebar

(d) search_fields configuration - enables keyword search across user IDs and pages

Working: Once registered, Django automatically generates a full set of list, detail, add, and delete views for the Event model — requiring no additional HTML or view code. This module proved particularly useful during development for manually verifying that events generated by the test script and demo storefront were being stored correctly.

5.5 Real-Time Signal Module (signals.py)

• **Functionality:** Automatically detects when a new event is saved to the database and triggers the real-time notification process. This is the core "trigger" of the entire live-update pipeline, and the single most important module for the system's real-time behaviour.Components:

• **Components:**

(a) post_save receiver - a function decorated with @receiver(post_save, sender=Event), which Django automatically calls every time an Event is saved

(b) created flag check - ensures the notification logic only runs for newly created events, not for edits to existing records

(c) Payload builder - converts the saved event's fields into a JSON-ready Python dictionary, converting the timestamp into local IST time using timezone.localtime()

(d)Channel layer dispatch - sends the constructed payload to Redis via channel_layer.group_send(), addressed to the group dashboard_updates

(e) async_to_sync bridge - allows this synchronous signal handler to safely call into the asynchronous Channels/Redis layer

Working: This module requires no manual invocation anywhere in the codebase Django's signal framework calls it automatically as a side effect of saving any Event, whether that save originates from track_event, the Django Admin panel, or the generate_data.py script. This design keeps the "what happens after save" logic cleanly separated from the "how the event was created" logic.

5.6 WebSocket Consumer Module (consumers.py)

• **Functionality:** Manages the live, persistent connection between the server and each connected browser, ensuring real-time updates are delivered the moment they are broadcast by the signal module.

• **Components**:

(a) DashboardConsumer class - extends Django Channels' AsyncWebsocketConsumer

(b) connect() method - runs when a browser opens a WebSocket connection; registers that connection into the dashboard_updates group and accepts the connection

(c) disconnect() method - runs when a browser closes its connection or the tab is closed; removes the connection from the group to prevent sending to a dead socket

(d) send_update() method - automatically invoked whenever a message is sent to the group; forwards the event data to the specific browser as a JSON-encoded WebSocket message.

Working: Every browser that opens the dashboard establishes its own independent WebSocket connection, but all of them join the same dashboard_updates group. This means a single event, broadcast once through Redis, is automatically delivered to every open dashboard simultaneously without the server needing to track individual connections manually.

5.7. WebSocket Routing and Protocol Module (routing.py and asgi.py)

• **Functionality:** Configures the application to handle WebSocket protocol connections alongside standard HTTP requests, which is not something Django supports natively without Channels.

• **Components:**

(a)websocket_urlpatterns (in routing.py) - maps the URL path ws/dashboard/ to DashboardConsumer

(b) ProtocolTypeRouter (in asgi.py) - inspects each incoming connection and routes standard HTTP traffic through Django's normal request-handling path, while routing WebSocket traffic through Channels' routing system

(c) AuthMiddlewareStack - wraps the WebSocket routing to make Django's authentication and session data available within consumers, consistent with normal Django views.

Working: This module effectively runs "underneath" the rest of the application, invisible during normal use but essential to its function without it, the browser's WebSocket connection request would fail entirely, and the project would only support standard, refresh-based data fetching.

5.8. Frontend Dashboard Module (dashboard.html)

• **Functionality:** Presents captured and analyzed data to the user through an interactive, three-tab interface, updating live as new events arrive without any manual refresh.

• **Components:**

(a) Live Dashboard tab - displays summary cards (total events, unique users, purchases, top page), a live auto-scrolling event feed, and a Chart.js doughnut chart broken down by event type

(b) Historical Data tab - presents a searchable, sortable table of recent events fetched from /api/events/

(c) About/Stack tab - a short reference section summarizing the technologies used in the project

(d) WebSocket client script - opens a persistent connection to ws/dashboard/ on page load, and updates the DOM (cards, chart, and feed) every time a message is received

(e) Initial data-loading logic - calls /api/summary/ and /api/events/ once on page load, ensuring the dashboard is populated correctly even before any new live event arrives.

Working: The page combines two data-loading strategies: an initial "pull" (via fetch() calls on load) to establish a baseline view, and a continuous "push" (via the open WebSocket) to keep that view current thereafter this hybrid approach avoids showing a blank dashboard while still achieving true real-time behaviour for anything that happens after the page loads.

5.9. Demo Storefront Module (demo_site.html)

• **Functionality:** Simulates realistic user activity by generating genuine tracked events through normal e-commerce-style interactions, used to demonstrate and validate the complete real-time pipeline end-to-end during presentations and testing.

• **Components:**

(a) Product cards ("StyleNest" branding) - each with an "Add to Bag" button that triggers both a click and a subsequent purchase event

(b) Sign-up button - triggers a signup event

(c) Hero banner - triggers a page_view event when interacted with

(d) track() JavaScript function - constructs a JSON payload (user_id, event_type, page) and sends it to /api/track/ via the browser's fetch() API

(e) Toast notification component - provides visual confirmation on-screen that an event was successfully sent, without requiring the user to check the dashboard separately.

Working: This module has no direct connection to the dashboard module in code the two communicate exclusively through the shared backend, exactly as an independent, real client-facing website would. This design choice was intentional, to demonstrate that the analytics pipeline works generically for any source of events, not just a tightly-coupled test harness.

5.10. Data Simulation Module (generate_data.py)

• **Functionality**: A standalone script used during development and testing to bulk-generate realistic random events, without requiring repeated manual interaction with the demo storefront for every test run.

• **Components:**

(a) Django environment bootstrap - configures the script to run outside the normal manage.py command flow while still accessing Django's models

(b) Random event generator - creates events with randomized user IDs, event types, and page values drawn from representative sample lists

(c) Bulk creation loop - inserts a batch of 100 events directly into the database in a single execution, each of which independently triggers the real-time notification pipeline described in Section 4.5

Working: Because each call to Event.objects.create() inside this script still passes through the same post_save signal as any other event creation, running this script provides a convenient way to demonstrate the dashboard updating rapidly and repeatedly useful for testing the system's behaviour under a higher volume of near-simultaneous events.

CHAPTER 6

PRACTICAL IMPLEMENTATION & TECHNICAL SNAPSHOTS

6.1. Development Environment Setup

The development environment for this project was carefully configured to support Python-based backend development alongside real-time messaging infrastructure.

• Code Editor: Visual Studio Code was used as the primary development environment, chosen for its integrated terminal, Python extension support, and debugging tools.

• Virtual Environment: A Python virtual environment (venv) was created to isolate project dependencies, ensuring packages installed for this project did not conflict with other Python projects on the system.

• Core Dependencies Installed: Django, Django REST Framework, Django Channels, Daphne (ASGI server), Pandas, and the Redis Python client library all installed within the virtual environment using pip.

• Database Server: PostgreSQL was installed and configured locally as the primary relational database.

• Message Broker Setup: Since Redis does not officially support Windows, Memurai a Windows-compatible, Redis-protocol-compliant server was installed and run as a background service to fulfil this role.

• Version Control Practice: Project files were organized following Django's standard app structure, keeping models, views, and real-time logic cleanly separated for maintainability.

6.2. Database Implementation

Database implementation focused on translating the application's data requirements into a reliable, queryable PostgreSQL schema.

• Configuration: The DATABASES setting in settings.py was updated to connect Django to PostgreSQL, specifying the database name, user credentials, host, and port.

• Schema Definition: The Event model was defined in models.py using Django's ORM, specifying field types and constraints in Python rather than raw SQL.

• Migrations: The makemigrations command was used to generate migration files describing the required schema changes, and migrate was used to apply them, creating the actual Event table in PostgreSQL.

• Verification: The Django Admin panel was used to manually confirm that the table was created correctly and that records could be added, viewed, and queried as expected.

• Data Integrity: Field-level constraints (such as restricted choices for event_type) were enforced at the model level, ensuring only valid data could be stored.

6.3. Real-Time Pipeline Implementation

This was the most technically involved part of the project, requiring multiple components to be configured to work together correctly.

• Channels Integration: Django Channels was added to INSTALLED_APPS, extending Django's default HTTP-only request handling to support WebSocket connections.

• ASGI Configuration: The project's asgi.py file was updated to use a ProtocolTypeRouter, directing standard HTTP traffic through Django's normal path and WebSocket traffic through Channels' routing system.

• Channel Layer Setup: A CHANNEL_LAYERS setting was configured to use Redis as the backing message broker, enabling communication between the signal handler and connected WebSocket clients.

• Signal Implementation: The signals.py module was implemented using Django's post_save signal, automatically triggering a notification each time a new Event was created.

• Consumer Implementation: The consumers.py module was implemented to manage WebSocket connections, handling client connect, disconnect, and message-forwarding logic.

• ASGI Server: Daphne was used to run the application in place of Django's default development server, since the default server does not support WebSocket protocol handling.

6.4.API Implementation and Testing

REST API endpoints were implemented and tested to ensure both raw and aggregated data could be reliably retrieved and submitted.

• Raw Data Endpoint: The event_list view was implemented to return the most recent 500 events as JSON, and tested by generating sample data and verifying the returned structure in the browser.

• Analytics Endpoint: The analytics_summary view was implemented using Pandas to compute totals, unique user counts, and event-type breakdowns, with results manually cross-checked against raw data for accuracy.

• Event Submission Endpoint: The track_event view was implemented to accept POST requests, tested directly through the demo storefront to confirm that submitted events were correctly parsed and saved.

• Error Handling: Basic checks were added to ensure malformed or non-POST requests to track_event returned an appropriate error response rather than failing silently.

• CORS Consideration: Since the demo storefront and dashboard are served from the same Django application, no cross-origin configuration was required, simplifying the integration

6.5. Frontend Dashboard Implementation

The dashboard frontend was built to prioritize clarity and responsiveness to live data, without introducing unnecessary framework complexity.

• Technology Choice: Built using vanilla HTML, CSS, and JavaScript with the Chart.js library, avoiding the overhead of a full frontend framework for a project of this scope.

• Initial Data Load: On page load, the dashboard calls the summary and events APIs to populate an initial snapshot of data before any live event arrives.

• WebSocket Connection: A persistent WebSocket connection is opened to ws/dashboard/, kept alive for the duration of the page session.

• Live UI Updates: Incoming WebSocket messages update three interface areas simultaneously summary statistic cards, the Chart.js doughnut chart, and a scrolling live event feed via direct DOM manipulation.

• Three-Tab Structure: The interface was organized into Live Dashboard, Historical Data, and About/Stack tabs, allowing users to switch between real-time and historical views without navigating to a different page.

System Architecture and Workflow Diagrams

This section presents the core technical diagrams describing how the system was built and how it operates.

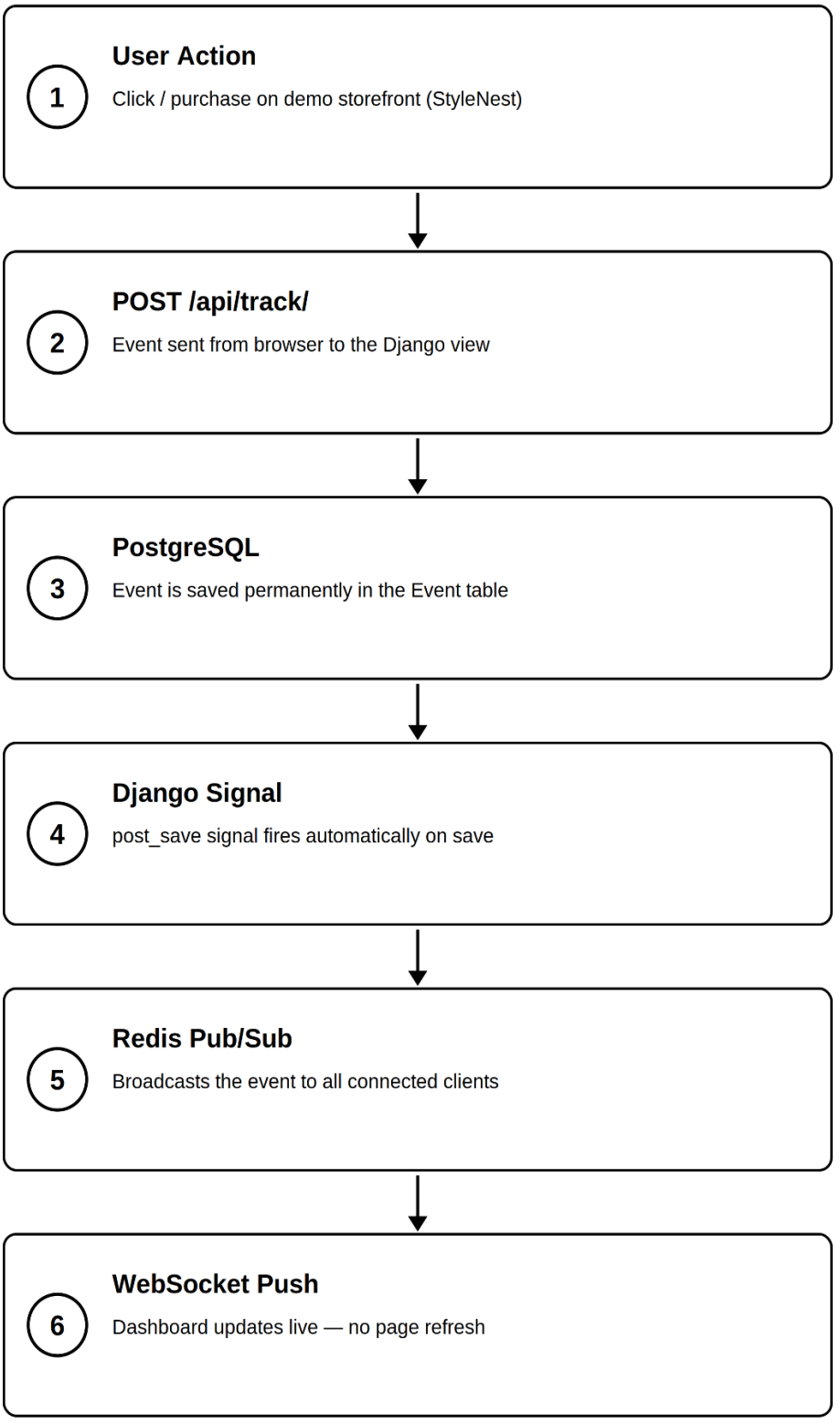

Figure 6.6. Real-Time Data Flow Architecture

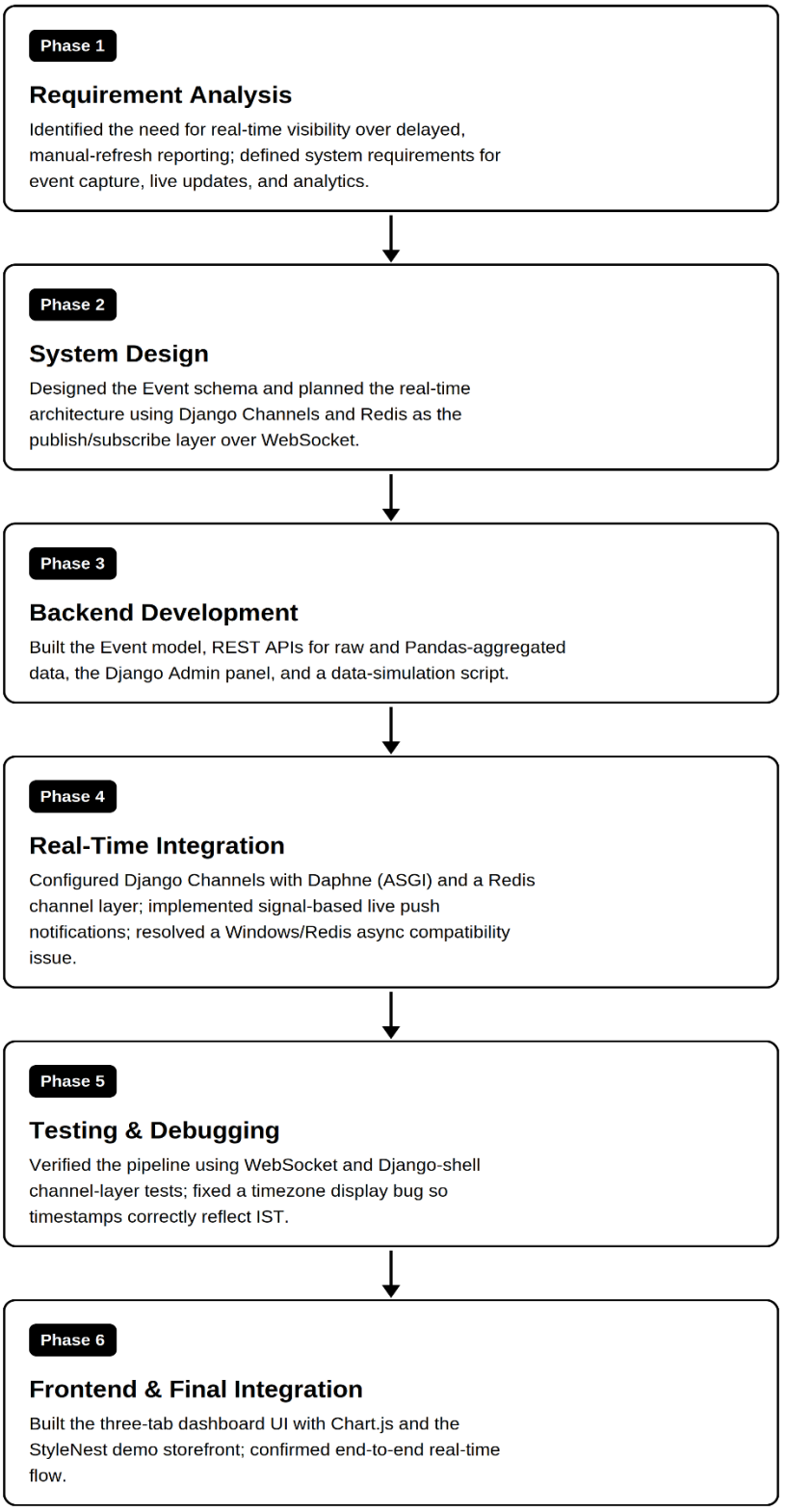

Figure 6.7. Software Development Life Cycle Workflow

Figure 6.8. Live Dashboard Interface

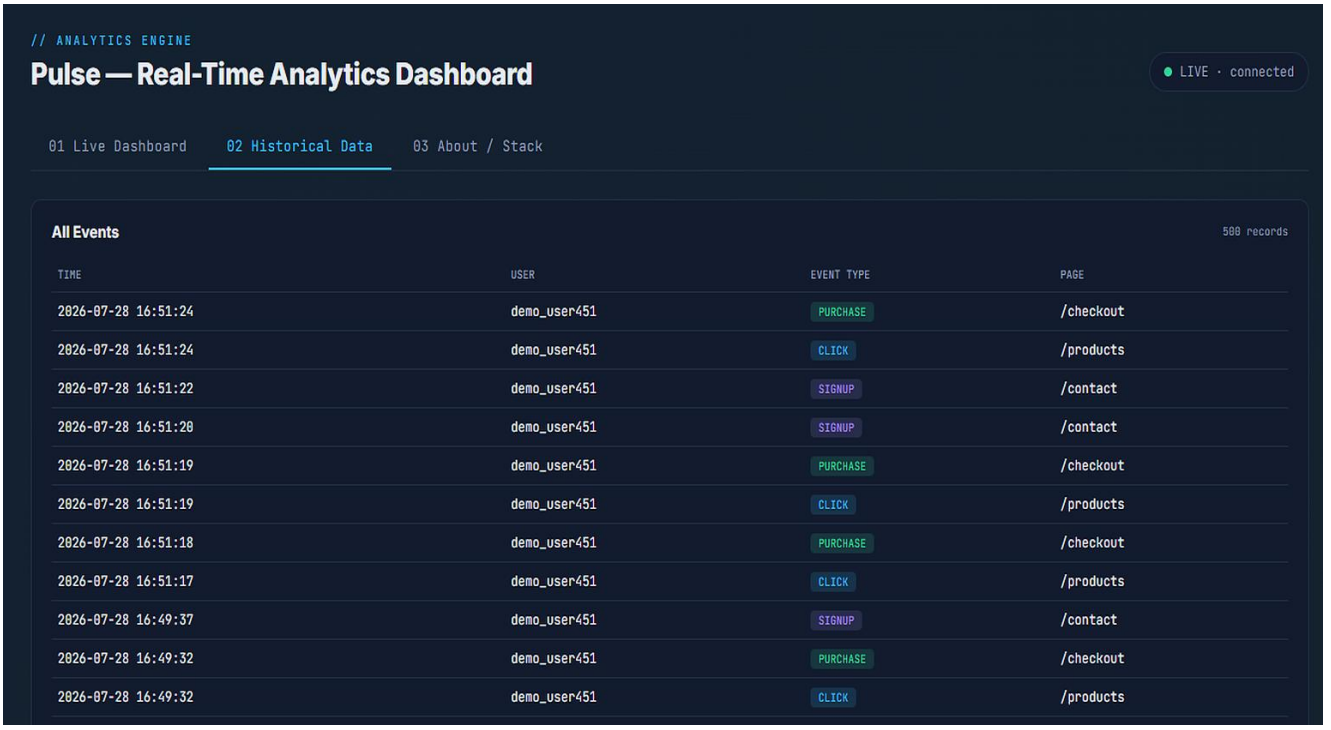

Figure 6.9. Historical Data Interface

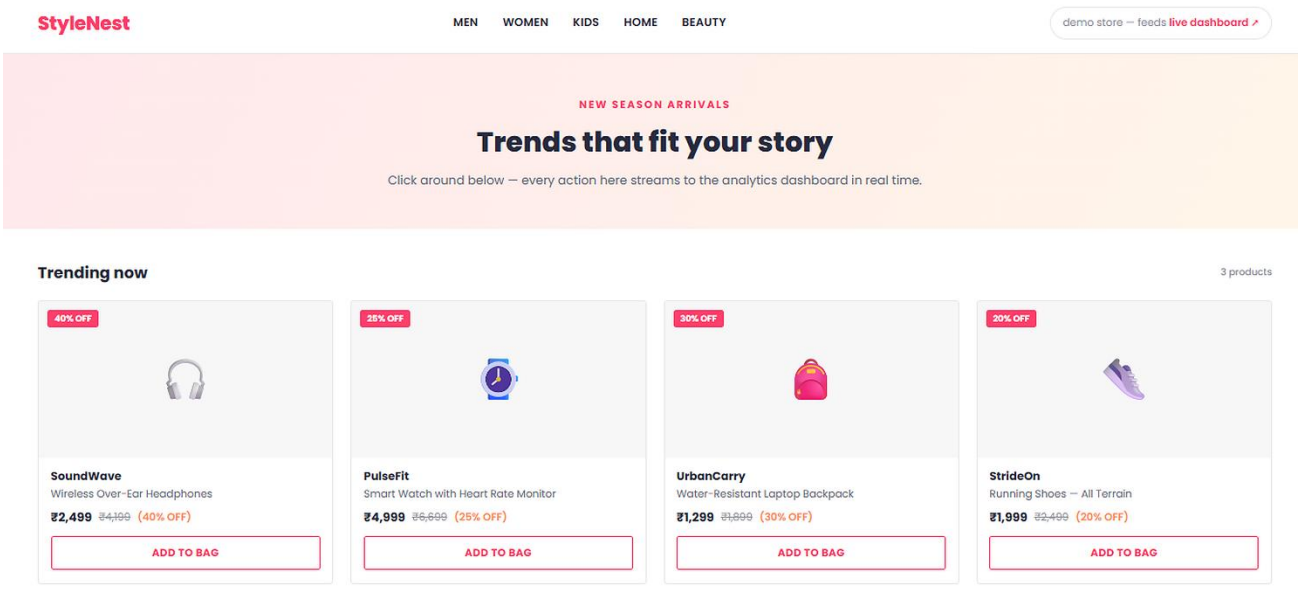

Figure 6.10. Demo Storefront Interface

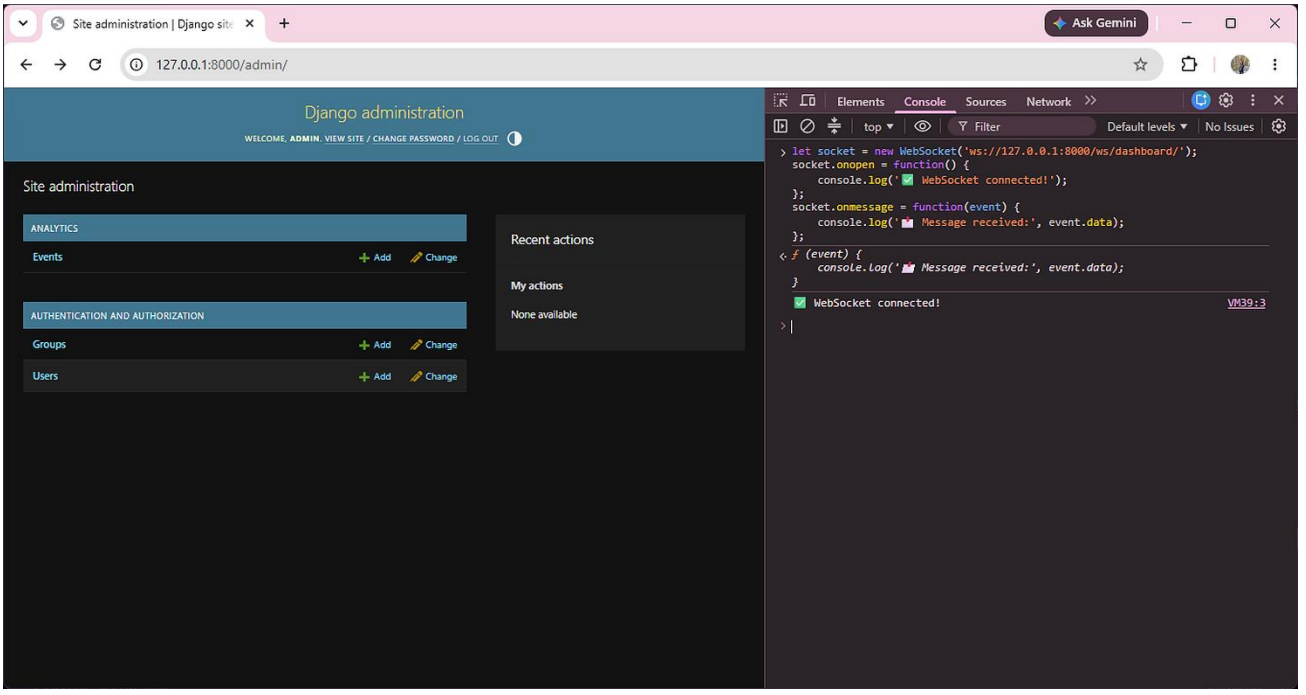

Figure 6.11. Django Administration Panel

CHAPTER 7

TRAINING OUTCOME

Pulse — Real-Time Analytics Dashboard is a full-stack real-time analytics application built using Python and Django, incorporating PostgreSQL, Django Channels, Redis, and Pandas. This project demonstrates the effective integration of these technologies to achieve reliable event capture, live data processing, and real-time visualization. The application includes advanced features such as WebSocket-based live updates, Pandas-driven data aggregation, and a companion demo storefront, showcasing a complete, production-style real-time data pipeline from event generation to live dashboard display.

7.1 Overview of the Industrial Training

During my industrial training with S O Infotech (P) Ltd., I received comprehensive, hands-on instruction in backend web development using the Python and Django ecosystem, extending into real-time systems architecture involving WebSockets and message-broker technologies. The training provided a deep dive into both conventional request-response backend development and the more advanced domain of asynchronous, event-driven programming, enabling me to master technologies and concepts that go well beyond typical academic coursework.

As part of the training, I applied my skills by developing "Pulse," a real-time analytics dashboard capable of capturing live user activity, processing it into meaningful statistics, and visualizing it instantly without manual refresh. This project involved building a complete data pipeline — from database design and REST API development, through real-time messaging infrastructure, to an interactive, live-updating frontend dashboard and a companion demo e-commerce storefront used to simulate realistic traffic.

- The training combined structured guidance with independent problem-solving, closely mirroring how real engineering teams operate.
- Weekly progress was structured around incremental milestones — starting with a working backend, then layering in real-time capability, and finally the presentation layer.
- Emphasis was placed on understanding *why* each technology choice was made, not just how to implement it.

7.2 Skills Acquired during Training

- **Backend Development:** Proficiency in building server-side applications using Django, including models, views, URL routing, and the Django Admin interface.
- **Database Management**: Expertise in designing and managing a relational schema in PostgreSQL, including field-level validation, choice constraints, and using Django's ORM and migration system to manage schema changes without writing raw SQL.
- **Real-Time Systems Development:** Hands-on experience implementing WebSocket-based communication using Django Channels, including consumer classes, group-based message broadcasting, and ASGI server configuration via Daphne.
- **API Development**: Designing and implementing RESTful JSON endpoints for both data retrieval and data submission, including handling POST requests from an external-facing client.
- **Message Broker Integration**: Practical understanding of Redis as a publish/subscribe message broker, and how it decouples event producers from event consumers in a scalable architecture**.**
- **Data Analysis with Pandas:** Ability to transform raw, row-level database records into aggregated, decision-ready statistics using DataFrame operations such as value_counts() and nunique().
- **Frontend Integration:** Building a responsive, live-updating dashboard interface using HTML, CSS, JavaScript, and Chart.js, and connecting it to backend data using both standard fetch() calls and persistent WebSocket connections.
- **Version Control and Environment Management:** Working with Python virtual environments to isolate dependencies, and structuring a Django project according to standard, maintainable conventions.
- **Debugging Complex, Cross-Platform Issues:** Diagnosing and resolving a subtle asynchronous compatibility issue between the Redis client library and the Windows operating system, through systematic, evidence-based troubleshooting.

7.3 Practical Application of Skills

The skills gained during this internship with S O Infotech are directly applicable to real-world software development. Backend development supports the construction of complete, functioning web applications from the ground up. Database management ensures data is stored efficiently, safely, and in a structurally sound way that supports future growth. Real-time systems knowledge enables the development of modern, responsive applications — such as live dashboards, chat systems, or notification services — that are increasingly expected in production software. Message broker integration supports building systems that scale beyond a single server or a single user. Data analysis skills allow raw operational data to be turned into actionable business insight. API development enables clean separation between frontend and backend, allowing each to evolve independently. Frontend integration ensures that all of this backend capability is actually usable by an end user. Together, these skills form a coherent, end-to-end capability: the ability to take a business problem, design a system to solve it, and implement that system from database to browser.

7.4 Enhanced Understanding of MERN

The internship with S O Infotech significantly enhanced my understanding of real-time web architecture, an area typically underexplored in standard academic curricula. I gained practical experience integrating PostgreSQL for structured, reliable data storage; Django Channels for extending a traditional web framework to support persistent WebSocket connections; Redis for efficient, scalable message broadcasting; and Pandas for meaningful data aggregation. This comprehensive exposure deepened my ability to reason about system design holistically — understanding not just how to write code for a single feature, but how multiple services (a web server, a database, and a message broker) must be coordinated to behave correctly together. I also developed a clear mental model of the distinction between synchronous and asynchronous execution, and why real-time systems require careful handling of this distinction — a concept directly reinforced through the Windows/Redis compatibility issue encountered and resolved during development.

7.5 Exposure to Industry Standards & Best Practices

The internship provided valuable exposure to industry standards and best practices in professional software development.

- I learned to write modular, maintainable code, with each Django file (models.py, views.py, signals.py, consumers.py) handling a single, clearly defined responsibility.
- I adopted the practice of testing components in isolation — for example, verifying the Redis channel layer directly through the Django shell before integrating it into the full signal-to-WebSocket pipeline.
- I learned the importance of environment isolation using Python virtual environments, preventing dependency conflicts between projects.
- Emphasis was placed on designing for scalability from the outset — using a message broker rather than direct database polling, so the architecture could support multiple simultaneous dashboard viewers without additional rework.
- I gained an appreciation for the value of clear internal documentation and code comments, particularly for the less intuitive real-time components, to support future maintainability.

This experience helped align my technical habits with the standards expected in professional, production-oriented software development.

7.6 Improvement in Problem-Solving Abilities

The internship significantly enhanced my problem-solving skills, particularly through the process of diagnosing a genuinely difficult, non-obvious technical issue. When WebSocket connections began silently timing out due to a Redis compatibility issue on Windows, I could not rely on a simple, well-documented fix. Instead, I developed a systematic debugging approach:

- Isolating the problem by testing each layer independently — first confirming the WebSocket connection itself worked, then testing the Redis channel layer directly via the Django shell, separate from the signal and consumer logic.
- Adding targeted debug output at each stage of the connection lifecycle to pinpoint exactly where the failure occurred.
- Reading and interpreting Python tracebacks carefully to distinguish the symptom (a timeout) from the underlying cause (an asynchronous I/O incompatibility).
- Researching and testing a targeted fix (pinning a specific package version) rather than making broad, speculative changes.

This process sharpened my analytical thinking and gave me a repeatable method for approaching unfamiliar technical problems — a skill far more valuable than memorizing any single fix.

7.7 Effective Team Collaboration & Communication

A significant part of this internship involved learning to explain a non-trivial technical system clearly both in writing (through this report) and verbally (through project discussions and presentation preparation). Translating a real-time architecture involving signals, message brokers, and WebSockets into language that is understandable to someone unfamiliar with the implementation required distilling complex mechanisms into clear, accurate explanations without oversimplifying to the point of inaccuracy. This process reinforced the understanding that technical competence is only fully valuable when it can be communicated whether to a teammate reviewing code, a mentor evaluating progress, or an evaluator assessing the completed project.

7.8 Introduction to Real-World Software Development

The combination of hands-on implementation and guided problem-solving provided an introductory but genuine view into the full software development life cycle. From the careful analysis of requirements identifying precisely why real-time visibility matters over delayed reporting through system design, backend implementation, real-time integration, testing, and final frontend delivery, the internship offered a structured path through each phase a real engineering project passes through. Rather than working from a fixed, pre-written specification, I experienced firsthand how requirements translate into architectural decisions, how those decisions surface unexpected technical challenges, and how a working system gradually emerges through iteration - providing an invaluable, practical complement to academic study of the software development life cycle.

7.9 Project Management &Time Management Skills

During the development of Pulse, deliberate project management and time management practices were applied to ensure steady progress within the internship timeframe.

- Work was broken into clear phases — environment setup, data modeling, API development, real-time integration, frontend development, and final testing — allowing progress to be tracked incrementally rather than attempted all at once.

- Tasks were prioritized based on dependency: for instance, ensuring the database and basic APIs were solid before attempting the more complex real-time layer, since later work depended directly on earlier components functioning correctly.
- Time was deliberately reserved for debugging, recognizing that unexpected technical issues (such as the Redis compatibility problem) are a normal and expected part of real development timelines, not a deviation from them.
- Regular self-review checkpoints were used to confirm that each completed phase genuinely worked end-to-end before moving to the next, reducing the risk of compounding errors later in the project.

CHAPTER 8
CONCLUSION

This report has presented the design, development, and implementation of Pulse a Real-Time Analytics Dashboard undertaken as part of an industrial training internship at S O Infotech (P) Ltd. The project set out to address a genuine and common limitation in conventional business dashboards: the delay between an event occurring and that event becoming visible to the people who need to act on it. Traditional systems, reliant on manual refresh or periodic batch reporting, inherently lag behind the reality they are meant to represent, and in fast-moving business contexts a sudden spike in traffic, a checkout process that has begun failing, a promotional campaign that is underperforming that lag can translate directly into missed opportunities or delayed responses to genuine problems. Pulse was built specifically to close that gap, using a combination of a relational database, a Python-based backend framework, an in-memory message broker, and a persistent WebSocket connection to ensure that data becomes visible on a live dashboard within roughly one second of being generated, with no manual intervention required at any point in that chain.

Over the course of this internship, the project progressed through a complete and structured development life cycle: beginning with requirement analysis and system design, moving through backend and database implementation, advancing into the more technically demanding real-time integration phase, and concluding with rigorous testing, debugging, and final frontend delivery. Each of these phases has been documented in detail in the preceding chapters, alongside the specific technologies employed, the architectural reasoning behind each decision, and the practical challenges encountered along the way. What began as a conceptual requirement visibility into user activity without delay was, by the end of the internship, a fully functioning system that could be demonstrated live, end to end, from a single click on a demo storefront to a corresponding visual update on a dashboard a fraction of a second later.

Reviewing the objectives set out in Chapter 4, the completed system can be evaluated point by point against what was originally intended, and in every case, the outcome met or exceeded that intention. A backend capable of capturing and processing user-activity events as they occur was successfully implemented using Django and PostgreSQL, with a well-structured Event model supporting reliable, validated data storage. Live visualization without manual refresh was achieved through the integration of Django Channels, Redis, and a WebSocket-based frontend, allowing the dashboard to update instantly whenever new data is recorded, without the user ever needing to press a refresh button or wait for a

scheduled report. Pandas was successfully applied to convert raw, row-level event data into meaningful aggregated analytics, including total event counts, unique user counts, event-type breakdowns, and top-page rankings, transforming what would otherwise be an undifferentiated stream of database rows into genuinely usable business insight.

A genuine push-based architecture was implemented in place of repeated polling, ensuring the system remains efficient even as the number of connected dashboard viewers grows, since Redis rather than the database itself bears the load of notifying multiple simultaneous clients. Realistic traffic simulation was achieved through the companion "StyleNest" demo storefront, which validated the complete pipeline under conditions closely resembling real user behaviour rather than relying on synthetic test data alone; this was, in many respects, the most convincing proof that the system worked as intended, since it demonstrated the pipeline functioning correctly with a completely independent, loosely-coupled source of events. Finally, the resulting system reflects patterns genuinely used in production-grade analytics platforms: a decoupled messaging layer, a clearly separated data and presentation tier, and a scalable, database-backed foundation. Taken together, every objective defined at the outset of this project was met, and in several respects particularly the depth of the real-time architecture and the inclusion of a working demo storefront the final system exceeded the project's original scope.

Key Outcomes and Learning

Beyond the technical deliverable itself, this internship produced outcomes that extend well beyond the boundaries of the project. Building a genuinely real-time system required moving past familiar, synchronous request-response thinking and developing a working understanding of asynchronous, event-driven architecture, a paradigm shift that is often absent from standard academic instruction but is increasingly central to modern software engineering. Concepts that had previously existed only as abstract terminology publish/subscribe messaging, event-driven signals, persistent socket connections became concrete, working parts of a system I had built and could reason about with confidence.

Encountering and resolving a subtle, platform-specific compatibility issue between the Redis client library and the Windows operating system proved to be one of the most valuable experiences of the internship, not because of the specific fix itself, but because of the systematic, evidence-based debugging process it demanded. Rather than guessing at solutions, the issue required isolating each layer of the system in turn, reading technical error output carefully, and distinguishing between a symptom and its underlying cause. This experience reinforced that professional software development is defined less by writing code that works on the first attempt, and more by the ability to methodically diagnose and resolve code that does not a lesson that will likely prove more durable than any single technical fact learned

during this project.

Equally significant was the experience of communicating a non-trivial technical system clearly, first through this written report, and subsequently through project presentation and viva discussion. Translating an architecture involving database signals, message brokers, and persistent socket connections into language accessible to an evaluator unfamiliar with the implementation required a genuine, internalized understanding of the system, rather than a superficial or memorized one. This process has left me confident not only in having built the system, but in being able to explain, justify, and defend every design decision within it from why PostgreSQL was chosen over a simpler alternative, to why a message broker was necessary at all, to what exactly happens in the fraction of a second between a user's click and a dashboard's update.

While the project successfully achieves its core objectives, certain areas remain open for future extension, consistent with the scope boundaries defined earlier in this report. The current system does not implement user authentication for dashboard access, which would be a necessary addition for any production deployment involving sensitive or client-specific data; at present, the dashboard is openly accessible to anyone with network access to the server, which is acceptable for a demonstration environment but would require revisiting before real-world use. The architecture, while scalable in principle, has not been load-tested beyond a single Redis instance and a moderate volume of simulated events, and a genuine production system would benefit from formal load testing, and potentially a distributed message-broker configuration capable of handling significantly higher event volumes and larger numbers of simultaneous dashboard viewers.

Future iterations of this project could also explore predictive analytics, such as anomaly detection on sudden traffic spikes or basic forecasting of expected activity levels, building directly on the event data already being captured without requiring any change to the underlying capture pipeline. Containerized deployment to a cloud platform would also be a natural next step, demonstrating the system operating outside a local development environment and closer to how it would genuinely be run in production. None of these limitations detract from what has been achieved within the scope of this internship; rather, they represent a realistic and honest acknowledgment of the difference between a well-executed academic project and a fully hardened production system, and they point toward a clear, achievable path should this project be extended further

.

This internship has been a genuinely transformative part of my academic journey, providing an

opportunity to move beyond theoretical coursework and construct a complete, working, real-time software system from first principles. The process of building Pulse, from an empty project folder to a fully integrated pipeline capable of capturing a live user action and reflecting it on a dashboard within a second, has given me practical, defensible expertise in backend development, real-time systems architecture, and data analysis that I expect to draw on throughout the remainder of my academic and professional career. It also gave me a realistic sense of what software engineering work actually involves day to day: not a smooth, linear progression from idea to finished product, but a process of steady progress interrupted by genuine, sometimes difficult problems, each of which had to be understood before it could be solved.

I am grateful to S O Infotech (P) Ltd. for the opportunity to undertake this training, and for the environment and guidance that allowed a project of this technical depth to be completed within the internship timeframe. I am equally grateful for the mentorship and support of my faculty guide at Shri Ram Murti Smarak College of Engineering & Technology, whose feedback helped shape both the project and this report into their final form. This experience has left me with a working system I can present with genuine understanding and confidence, and, more importantly, with a set of technical and problem-solving skills I am confident will extend well beyond this project alone.

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APPENDIX

Appendix A: Sample Code Listings

The following illustrative code excerpts represent the structure and logic of key Server Actions implemented during the internship. They are simplified for readability in this report; the full implementation includes additional error handling and validation.

A.1 models.py — Event Data Model

```
# analytics/models.py
from django.db import models

class Event(models.Model):
    EVENT_TYPES = [
        ("click", "Click"),
        ("purchase", "Purchase"),
        ("signup", "Signup"),
        ("page_view", "Page View"),
    ]
    user = models.CharField(max_length=100)
    event_type = models.CharField(max_length=20, choices=EVENT_TYPES)
    page = models.CharField(max_length=200)
    timestamp = models.DateTimeField(auto_now_add=True)

    class Meta:
        ordering = ["-timestamp"]
```

A.2 signals.py — Real-Time Signal Broadcast (simplified)

```
# analytics/signals.py
from django.db.models.signals import post_save
from django.dispatch import receiver
from channels.layers import get_channel_layer
from asgiref.sync import async_to_sync
from .models import Event

@receiver(post_save, sender=Event)
def broadcast_event(sender, instance, created, **kwargs):
    if not created:
        return
    channel_layer = get_channel_layer()
    async_to_sync(channel_layer.group_send)(
        "dashboard",
        {
            "type": "send_event",
            "data": {
                "user": instance.user,
                "event_type": instance.event_type,
                "page": instance.page,
                "timestamp": str(instance.timestamp),
            },
        },
    )
```

A.3 consumers.py — WebSocket Consumer (simplified)

```
# analytics/consumers.py
import json
from channels.generic.websocket import AsyncWebsocketConsumer

class DashboardConsumer(AsyncWebsocketConsumer):
    async def connect(self):
        await self.channel_layer.group_add("dashboard",
self.channel_name)
        await self.accept()

    async def disconnect(self, close_code):
        await self.channel_layer.group_discard("dashboard",
self.channel_name)

    async def send_event(self, event):
        await self.send(text_data=json.dumps(event["data"]))
```

Appendix B: Glossary of Terms

● API (Application Programming Interface): A defined set of rules that lets one piece of software communicate with another.

● ASGI (Asynchronous Server Gateway Interface): The async successor to WSGI, enabling Django to handle WebSockets alongside regular HTTP requests.

● Channel Layer: A Django Channels abstraction (backed by Redis) that lets different consumers/processes talk to each other and broadcast messages to groups.

● Consumer: The Channels equivalent of a Django view, but for handling WebSocket connections instead of HTTP requests.

● CRUD: Create, Read, Update, Delete the four basic operations of persistent data storage.

● Daphne: The ASGI application server used to run Django Channels in production/development.

● JSON (JavaScript Object Notation): A lightweight, text-based data format used for exchanging structured data between systems.

● ORM: Object-Relational Mapping a technique for converting data between incompatible type systems, commonly used to interact with relational databases from application code.

● Pub/Sub (Publish/Subscribe): A messaging pattern where a publisher broadcasts messages to a channel, and all subscribed clients receive them instantly used here via Redis.

● REST (Representational State Transfer): An architectural style for designing networked APIs using standard HTTP methods.

● Signal: A Django mechanism that lets certain senders notify a set of receivers when an action occurs (e.g., post_save firing after a model instance is saved).

● WebSocket: A communication protocol providing a persistent, full-duplex connection between client and server, enabling real-time data push without polling.

Appendix C: Supporting Documents

The following documents, referenced throughout this report, are attached as supporting evidence of the internship engagement:

- Offer Letter : issued by S O Infotech (P) Ltd., confirming the internship role, duration, and reporting structure.
- Completion Certificate : issued S O Infotech (P) Ltd., on completion of the internship tenure.

Appendix D: Environment Variables Reference

The following environment variables were required to run the application, split between the frontend and backend. Actual values are kept secret and are never committed to the source repository; only variable names and their purpose are documented here.

D.1 Django Settings

● SECRET_KEY — Django's cryptographic signing key used for sessions, tokens, and CSRF protection.

● DEBUG — Boolean flag controlling whether detailed error pages are shown; set to False in production.

● ALLOWED_HOSTS — List of domains/IP addresses permitted to serve the Django application.

D.2 Backend Environment Variables

● DATABASE_NAME / DATABASE_USER / DATABASE_PASSWORD / DATABASE_HOST / DATABASE_PORT — Connection parameters for the PostgreSQL database.

D.3 Real-Time / Channel Layer Configuration

● REDIS_HOST / REDIS_PORT — Address of the Redis instance used as the Channels channel layer for Pub/Sub messaging.

● CHANNEL_LAYERS_BACKEND — Backend class configured in settings.py (e.g., channels_redis.core.RedisChannelLayer) that connects Django Channels to Redis.